

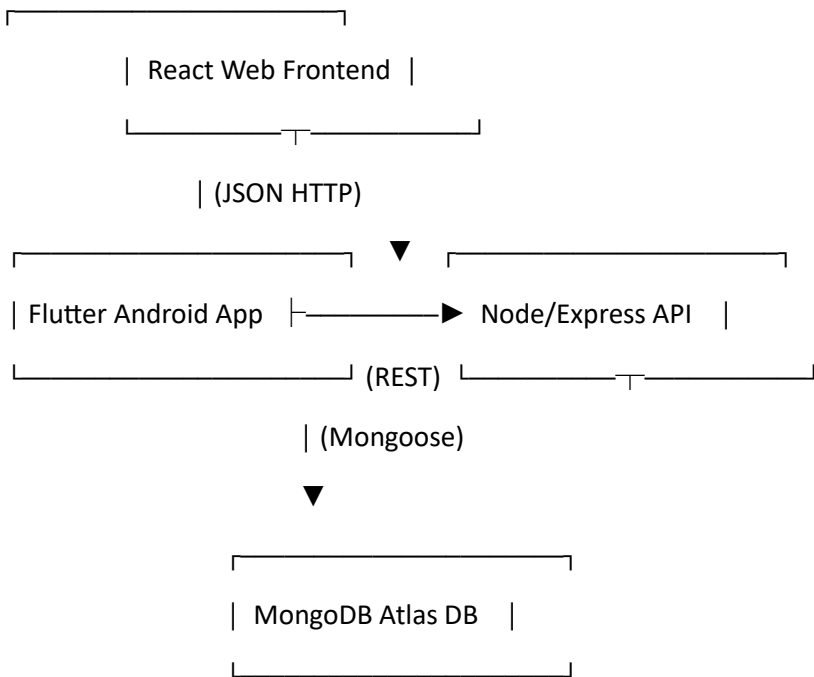
GuruBramha — Complete Flutter Developer Handover Document

Project: GuruBramha Academy App (Shared Web & App Platform)

Status: Ready for Integration

1 Project Overview & Shared Architecture

GuruBramha is a unified online learning and interactive coding platform. Both the **React Web App** and the **Flutter Mobile App** connect to the **same backend API and MongoDB database** to maintain absolute parity of users, courses, coding submissions, and leaderboards.



2 API Base URLs & Environment Setup

To establish secure communication from the Flutter application to the API, use the appropriate base URL configuration based on your runtime environment:

Environment	URL	Description / Notes
Local Dev (Android Emulator)	http://10.0.2.2:5000	Mandatory for Android emulators to route back to host's localhost:5000
Local Dev (Real Android Device)	http://<YOUR_PC_IP>:5000	Host and device must be on the same local Wi-Fi network
Production (Live)	https://gurubramha-api-hzbvbygvaawcpaueg.southindia-01.azurewebsites.net/	Live Azure server endpoint

3 Authentication APIs (Auth)

Base Path: /api/auth

- **Sign Up:** POST /api/auth/signup
 - *Body:* {"displayName": "...", "email": "...", "phone": "...", "address": "...", "password": "..."}
- **Login:** POST /api/auth/login
 - *Body:* {"email": "...", "password": "..."}
- **Forgot Password:** POST /api/auth/forgot-password
 - *Body:* {"email": "..."}
- **Reset Password:** POST /api/auth/reset-password/:token
 - *Body:* {"password": "..."}
- **Check Phone Uniqueness:** POST /api/auth/check-phone
 - *Body:* {"phone": "..."}

4 Problems & Practice APIs (Student + Admin CRUD)

Base Path: /api/problems

- **GET All Problems:** GET /api/problems (*Supports filters: searchQuery, difficulty, category, status, company, level*)
- **GET Single Problem:** GET /api/problems/:id (*Returns starter code, limits, hidden and public test cases*)
- **Admin Create:** POST /api/problems
- **Admin Update:** PUT /api/problems/:id
- **Admin Delete:** DELETE /api/problems/:id
- **Admin Duplicate:** POST /api/problems/:id/duplicate
- **Admin Reorder:** PUT /api/problems/reorder (*Body: {"orderedIds": [...]}*)
- **Admin Bulk Import:** POST /api/problems/import (*Body: {"problems": [...]}*)
- **Admin LeetCode Sync:** POST /api/problems/sync-leetcode
 - *Body:* {"syncUrl": "...", "limitCount": 100}

5 Code Sandbox & Execution APIs

Base Path: /api/submissions

- **Run Code (Quick Test):** POST /api/submissions/execute
 - *Body:* {"language": "python", "code": "...", "stdin": "..."}
 - **Submit Code (Full Evaluation):** POST /api/submissions/submit
 - *Body:* {"problemId": "...", "userId": "...", "language": "...", "code": "..."}
 - **Leaderboard:** GET /api/submissions/leaderboard
 - **Admin Submissions Log:** GET /api/submissions
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6 Courses APIs (Student + Admin Upload)

Base Path: /api/courses

- **GET All Courses:** GET /api/courses
 - **Admin Create:** POST /api/courses
 - **Admin Update:** PUT /api/courses/:id
 - **Admin Delete:** DELETE /api/courses/:id
 - **Admin File Upload:** POST /api/courses/upload (*multipart/form-data with file key, max 500MB*)
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7 Contests APIs (Student + Admin CRUD)

Base Path: /api/contests

- **GET All Contests:** GET /api/contests
 - **GET Single Contest:** GET /api/contests/:id
 - **Admin CRUD:** POST /api/contests, PUT /api/contests/:id, DELETE /api/contests/:id
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8 Documents APIs (Student + Admin CRUD)

Base Path: /api/documents

- **GET All Documents:** GET /api/documents
 - **Admin CRUD:** POST /api/documents, PUT /api/documents/:id, DELETE /api/documents/:id
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9 Razorpay Payment APIs

Base Path: /api/payment

- **Create Order:** POST /api/payment/create-order (*Amount in paise, e.g. ₹499.00 = 49900*)

- **Verify Payment Signature:** POST
/api/payment/verify (Body: razorpay_order_id, razorpay_payment_id, razorpay_signature)
 - **Razorpay SDK Key ID:**
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10 Supported Programming Languages

Use these exact keys inside the code execution

body: javascript, typescript, python, cpp, java, c, csharp, go, kotlin, sql, bash.

1 1 Dart Models (Copy-Paste Ready)

The flutter_developer_handover.md file includes fully typed, production-ready UserModel and ProblemModel Dart classes with fromJson and toJson implementations.

1 2 Required HTTP Headers

dart

```
Map<String, String> requestHeaders = {  
  'Content-Type': 'application/json',  
  'Accept': 'application/json',  
};
```

1 3 API Error Handling & Codes

All HTTP response failures adhere strictly to a standardized JSON error format: { "error": "Descriptive message here" }

1 4 Admin Access & Auto-Seeding Credentials

Field	Credentials / Login Value
Admin Email	adminguru@gmail.com
Admin Password	admin@1234
Admin Login Panel	http://localhost:5173/login

Auto-Seeding: Logging in with adminguru@gmail.com for the first time instructs the backend to **auto-create** and tag this account as an Administrator in MongoDB.

1 5 CRITICAL — What NOT to Share (Security Protocols)

- **MongoDB Atlas URI** (*Database connection string*)
 - **JWT Encryption Secret Key** (*Auth signing token*)
 - **Razorpay Key Secret** (*Backend payment signature verify*)
 - **Gmail SMTP Application Password** (*Backend email dispatcher*)
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1 6 Complete Handover Checklist

- **API Base URL (Dev):** `http://10.0.2.2:5000`
- **API Base URL (Prod):** `https://gurubramha-api-hzbvbygvcawcpaueg.southindia-01.azurewebsites.net/`
Razorpay Key ID: `rzp_test_Sr8E9gS2s7GAxp`
- **Admin Panel:** Protected via auto-seeded account credentials.
- **Database:** Shared MongoDB Atlas GuruBramha database.